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Hybrid Polarizing Solids with Extended Pore Diameters for Dissolution Dynamic Nuclear Polarization

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Dissolution dynamic nuclear polarization (dDNP) has emerged as a powerful technique to address the inherent sensitivity limitations in nuclear magnetic resonance imaging and spectroscopy. The technique relies on the use of stable paramagnetic polarizing agents that act as a source of hyperpolarization but unfortunately, their presence contaminates the sample and significantly enhances paramagnetic relaxation of the hyperpolarized solutions. In the last decade, novel sample formulations have been proposed to combat this issue, including silica-based hybrid polarizing solids (HYPSO), which have enabled the production of contamination-free hyperpolarization. Herein, a new generation of polarizing solids with pore diameters increased from 4 up to 28 nm is presented. Compared to the previously described HYPSOs with smaller pore sizes of 4 nm, larger pores ultimately achieve the ambitious contradictory goals of 1) providing rapid and high solid-state polarization and improving cross-polarization efficiency, while 2) decreasing hyperpolarization losses upon melting, transfer, and filtration, and 3) reducing the overall concentration of the polarizing agent by a factor of 2. These HYPSO materials can be used in virtually any existing dDNP settings without modification, apart from the sole addition of an in-line commercial filter to retain the HYPSO powder upon dissolution and transfer.

1. Introduction

Dissolution dynamic nuclear polarization (dDNP)[1] has established itself as a highly efficient method to overcome nuclear magnetic resonance imaging and spectroscopy (MRI and NMR) lack of sensitivity. By leveraging hyperpolarization,[2] dDNP enhances the nuclear spin polarization by several orders of magnitude, thereby greatly improving the sensitivity. This method involves microwave-induced transfer of high electron spin polarization, usually from free radicals comixed with the sample, to the nuclear spins of interest in a frozen state at low temperatures (typ. 1.2–1.8 K), and high magnetic fields (typ. 3.35–7 T). Subsequently, a controlled and irreversible process of melting, dilution, and transfer is executed, enabling the hyperpolarized sample to be analyzed using a room-temperature liquid-state NMR apparatus.[3] In such a setup, the free radicals necessary for DNP are inevitably dissolved together with the molecules of interest during dissolution, resulting in final radical concentrations (typ. $\approx$ 1 mM) that bring undesirable paramagnetic relaxation in the liquid state, leading to line broadening and hyperpolarization losses.[4]

In the last decade, novel sample formulations have been proposed[5] to combat this issue. One of these formulations involves the physical separation of the polarizing agents from the molecules of interest in the solid-state sample to be polarized. To achieve this goal, solid hyperpolarizing matrices containing covalently immobilized stable radicals are impregnated with the solutions that need to be polarized, followed by freezing in the polarizer, microwave-induced DNP, and filtration during the dissolution and transfer process. Polymers have emerged as promising polarizing media in recent years, initially pioneered by Prof. S. Han[6] and later by Prof. K. Münnemann,[7] specifically for liquid DNP. In these studies, nitroxides were covalently linked to Sepharose beads in the former case and to thermoresponsive gels in the latter. These amorphous polymers were subsequently used as polarizing agents for dDNP.[8]

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Alternatively, we introduced the utilization of mesostructured or nonstructured mesoporous silica matrices for dDNP, known as hybrid polarizing solids (HYPSO),[9–11] which provided enough ^{13}C liquid-state hyperpolarization after filtration to be compatible with imaging applications.[12] In this case, a silica layer containing nitroxide radicals was coated onto the surface of mesoporous silica beads exhibiting pore diameters of 4–5 nm. The resulting solids were irradiated, first with frequency-modulated[13] microwaves at high field[14] (6.7 or 7.05 T) to polarize the ^{1}H spins of the partly protonated solvent, and second with radiofrequency waves to subsequently transfer their polarization to ^{13}C spins of the protonated or deuterated[15] molecules of interest through microwave-gated[16] cross polarization (CP).[17–19] These HYPSO materials also recently proved to be efficient polarizing agents even in the context of benchtop DNP at 1 T and 77 K.[20,21] All the previously cited classes of HYPSO polarizing matrices featured pore diameters of up to 6 nm, which raised serious challenges regarding relaxation during the dissolution process due to the proximity of the hyperpolarized molecules with the polarizing agents located at the surface of the pores. Indeed, hyperpolarized molecules must "escape" the paramagnetic pores during the dissolution process as rapidly as possible to better preserve polarization in the liquid state. We previously observed that HYPSO 5 matrices exhibiting highly interconnected pores with random orientations[12] were superior in terms of the resulting liquid-state polarization compared to materials with well-aligned linear 2D hexagonal[9] (SBA-15 based HYPSOs) or 3D cubic[11] (SBA-16 based HYPSOs) pore-networks.

More recently, with the goal of generating very long-lived ^{13}C hyperpolarization and performing hyperpolarization transport,[22,23] we have made further progress by incorporating nitroxide radicals into rigid and porous epoxy polymer matrices, referred to as hyperpolarizing polymers (HYPOPs).[24] These HYPOPs display pore diameters ranging from hundreds of nanometers to several micrometers, which on the one hand confers very long lifetimes to the hyperpolarized ^{13}C spins (due to their distances to the paramagnetic nitroxides), but on the other hand, leads to an extension of the polarization build-up times that are limited by ^{1}H spin diffusion. This observation led us to realize that larger pores are not only highly compatible with dDNP but could also provide unprecedented performance in sustaining ^{13}C hyperpolarization. This is beneficial both in the solid-state, for improved hyperpolarization transport, and in the liquid-state by reducing hyperpolarization during melting and dissolution.

We investigate here the feasibility of dDNP using HYPSO matrices with interconnected pores and larger pore dimensions as the best compromise for both optimal DNP performances and reduced losses upon dissolution, filtration, and transfer. We introduce a new generation of mesoporous silica matrices exhibiting 12 and 28 nm pore diameters, using the **HYPSO 5**[12] synthetic approach, and find that DNP performances are preserved at least for pores diameters in that range and that the CP efficiency to ^{13}C is improved with the increase of the pore diameter. We also observed that HYPSOs with larger pores require less radical concentration, which is a key advantage for dissolution as it leads to reduced paramagnetic relaxation during melting. These are indeed critical times during which molecules escaping pores are potentially subjected to exacerbated relaxation, as demonstrated by T_2 relaxation exchange spectroscopy (REXSY).[25,26] We also implement the use of a commercial and robust filtering solution to readily generate pure hyperpolarized solutions.

2. Results and Discussion

2.1. HYPSO 5 Summary

HYPSO 5 are hybrid polarizing solids based on commercially available porous silica spheres coated with a silica layer containing a polarizing agent (in this case, nitroxide radicals). **HYPSO 5** can withstand cryogenic temperatures down to superfluid helium (≈1 K) as well as relatively high temperatures up to 343 K. They feature substantial porosity with pore volumes higher than 0.7 $cm^3\,g^{-1}$ and mean pore sizes from 4–5 nm (in our previous study[12]) to 12 and 28 nm in this study. The concentration of the polarizing agent embedded in the silica layer can be tuned to a concentration suitable for optimal DNP (typically in the range of 10 to 100 $\mu mol\,g^{-1}$).

2.2. HYPSO 5 Synthesis

Two families of HYPSO materials with two different pore sizes (12 and 28 nm) and a broad range of TEMPO radical concentrations (from 9 to 67 $\mu mol\,cm^{-3}$) were prepared using a two-step procedure as previously described by our group in 2018.[12] These two **HYPSO 5** families are referred to as **HYPSO 5-12** and **HYPSO 5-28**. The first step of the preparation consisted of coating commercially available silica spheres (SiliCycles) with a homogeneous silica layer containing azidopropyl fragments, termed **N_3-silica spheres**. This coating was carried out by the dropwise addition of a THF solution of tetraethylorthosilicate and (3-azidopropyl) triethoxysilane at room temperature for 20 min on a preheated (70 °C) THF/HCl suspension of silica spheres. The resulting mixture was stirred under reflux for 1 h. The as-obtained solid was further recovered by filtration; washed with water/THF (20:80), EtOH, and Et_2O; and dried under high vacuum (10^{-5} mBar) at 135 °C overnight (see the Supporting Information for a detailed description of the experimental protocol).

The **N_3-silica spheres** were further transformed into the targeted **HYPSO-5** solids containing the TEMPO radicals through copper-catalyzed azide-alkyne cycloaddition (Cu-AAC). The reaction was performed under an Ar atmosphere by the addition of o-propargyl TEMPO to a DMF suspension of the previously prepared **N_3-silica spheres** and the further addition of a DMF solution of CuI. The resulting mixture was stirred for 72 h at 50 °C, and the resulting solid was collected by filtration; washed with DMF, EtOH, and Et_2O; and dried under vacuum (10^{-5} mbar) at 50 °C for 15 h (see the Supporting Information for a detailed description of the experimental protocol). The resulting solids were characterized by N_2 adsorption–desorption at 77 K, and the radical concentrations were measured by quantitative X-band EPR (see **Table 1** and Table S2, Supporting Information).

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Table 1. Texture parameters of HYPSO 5 samples, obtained from N_2 adsorption/desorption and EPR: surface area (S_{BET}), mean pore diameter (D_p), total pore volume (V_p(tot)), and radical concentration ([*R*]). All materials displayed typical type IV N_2 adsorption/desorption isotherms for mesoporous materials.

Notation: HYPSO 5-	S_{BET}[a] [$m^2 g^{-1}$]	D_p[b] [nm]	V_p(tot)[c] [$cm^3 g^{-1}$]	[*R*][d] [$\mu mol\ cm^{-3}$]
4-34[e]	447	4	0.64	34
–[f]	290	12	0.96	0
12-20	290	12	0.81	20
12-31	302	11	0.84	31
12-51	311	11	0.88	51
12-67	316	11	0.87	67
–[f]	110	28	0.78	0
28-9	98	26	0.70	9
28-13	106	28	0.76	13
28-18	116	28	0.75	18
28-27	100	28	0.74	27

[a]Calculated using the Brunauer–Emmett–Teller equation.[43] [b]Measured using the Barrett-Joyner-Halenda model[44] on the desorption branch of the N_2 adsorption/desorption isotherm. [c]Total pore volume corresponding to the quantity of N_2 adsorbed at $P/P_0 = 0.99$. [d]TEMPO concentrations in $\mu mol\ cm^{-3}$ correspond to the TEMPO concentrations in $\mu mol\ g^{-1}$, determined by EPR measurements on the final HYPSO solids, divided by HYPSO solids porous volume (in $cm^3 g^{-1}$). See Table S2, Supporting Information, for more information. [e]For the preparation procedure and detailed characterization for this sample, see ref. [12] [f]Pristine silica spheres.

2.3. DNP Performances of HYPSO 5-12 and HYPSO 5-28

Figure 1 shows the maximum ^1H polarization $P(^1H)$ reached at $T = 1.6$ K and $B_0 = 7.05$ T with DNP (see Experimental Section) for each HYPSO 5-12 and HYPSO 5-28 sample containing different radical concentrations and impregnated with H_2O/D_2O solutions (1/9 v/v), together with the associated DNP build-up rate $R_{DNP}(^1H) = 1/\tau_{DNP}(^1H)$ (where $\tau_{DNP}(^1H)$ is the ^1H DNP build-up time) and stretch parameter β.[3] The green square marks represent the results obtained for HYPSO 5-12 and the blue triangle marks correspond to the results obtained with HYPSO 5-28. The optimal ^1H polarization values were found to be $P(^1H) = 83\%$ and 71% at concentrations of 31 and 18 $\mu mol\ cm^{-3}$ for HYPSO 5-12 and HYPSO 5-28, respectively (hereinafter denoted as HYPSO 5-12-31 and HYPSO 5-28-18), with build-up rates $R_{DNP}(^1H) = 0.21$ and 0.12 min^{-1} and stretch parameters $\beta = 0.74$ and 0.81, respectively. Although the stretch parameter is similar for both optimal materials, the maximum polarization and build-up rate are only slightly degraded by a factor of 1.2 and 1.75 upon an increase in pore diameter by a factor of 2.3. The decrease in the DNP build-up rates for larger pores can be explained by a combination of two factors:

i) Optimal DNP occurs at a lower overall radical concentration, by a factor of 1.7 smaller for HYPSO 5-28 than for HYPSO 5-12, and lower radical concentrations generally lead to slower polarization dynamics in frozen solutions.

ii) Larger pores require prolonged ^1H nuclear spin diffusion over longer distances with a $\sqrt{Dt}$ dependence (where *D* is the diffusion coefficient of the nuclear spin, and *t* is time).

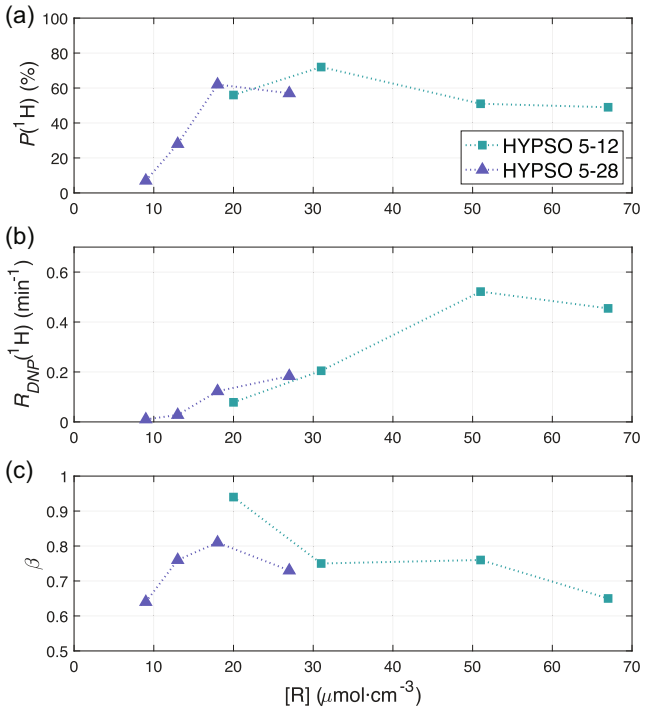


Figure 1. ^1H DNP performances of the HYPSO 5-12 series & HYPSO 5-28 pores series. The solids were impregnated with a solution of H_2O/D_2O (1/9 v/v), polarized at 1.6 K and 7.05 T (^1H nuclear Larmor frequency 300.14 MHz) (see the Experimental Section). a) Maximum ^1H polarization levels. The green squares (■) and blue triangles (▲) represent the values for HYPSO 5-12 and HYPSO 5-28 (12 and 28 nm pore sizes respectively). b) ^1H DNP rates, found by fitting polarization build-up curves with mono-exponential stretched curves. c) Stretch parameter β. Dashed lines, connecting experimental points, were added as a guideline to the eye.

Despite these two concurrent effects, the build-up rate remains astonishingly high for HYPSO 5-28 compared with the reported polarization build-up rates of conventional glassy frozen solutions in the literature[3] (generally the range 0.2–1 min^{-1}).

To investigate further the influence of pore size on DNP performance, these two optimal HYPSOs (HYPSO 5-12-31 and HYPSO 5-28-18) were selected for further comparison studies with HYPSO featuring smaller pore sizes. HYPSO 5 with mean pore size 4 nm and radical concentration 34 $\mu mol\ cm^{-3}$ (HYPSO 5-4-34) was previously reported to be a highly efficient polarizing matrix, with ^1H polarization values reaching near-unity.[12] Therefore, it was chosen as a third HYPSO for comparison. **Figure 2** shows the full ^1H DNP build-up curves for these three optimal HYPSOs at 1.6 K and 7.05 T, using 3 M sodium [1-^{13}C]-acetate in H_2O/D_2O (1/9 v/v) solution as a substrate.

Note that the absolute ^1H polarization values are not presented here, since proper quantification was impossible due to a very large radiation damping effect leading to broad lines and substantial signal loss during the spectrometer dead time.[3] Therefore, we decided here to compare different HYPSOs by using the ^1H NMR signal integral (normalized to the number of ^1H spins in each sample). As expected, ^1H polarization decreases with an increase in pore size and a decrease in the radical concentration, although the precise influence of these two parameters is ambiguous. For example, between HYPSO 5-12-31 and HYPSO 5-4-34, the radical concentration (per

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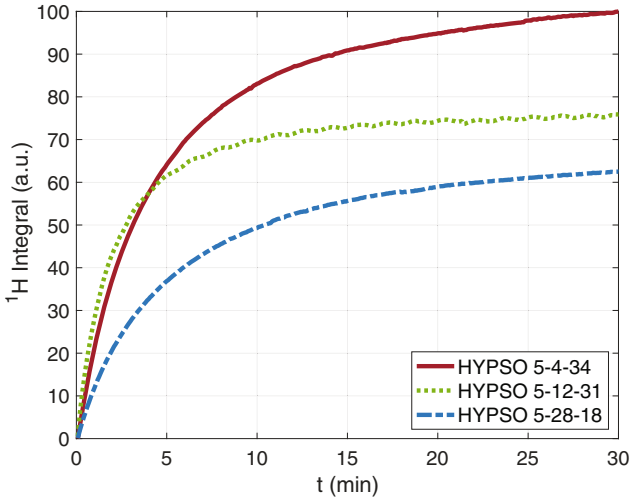


Figure 2. ^{1}H DNP build-up curves (i.e., ^{1}H NMR signal intensity, normalized by the amount of ^{1}H nuclei in each sample, presented on a relative scale) for the optimal HYPSO 5-4 at 34 μmol cm^{-3} radical concentration, HYPSO 5-12 at 31 μmol cm^{-3} radical concentration, and HYPSO 5-28 at 18 μmol cm^{-3} radical concentration, displayed with a dark red solid line, green dotted line, and blue dash-dotted line, respectively. The solids were impregnated with a 3 M sodium [1-^{13}C]-acetate in H_2O/D_2O (1/9 v/v) solution and polarized at 1.6 K and 7.05 T (see the Experimental Section). NMR was acquired every 5 s with 64 transients (*rf*-pulse flip-angle = 0.1°).

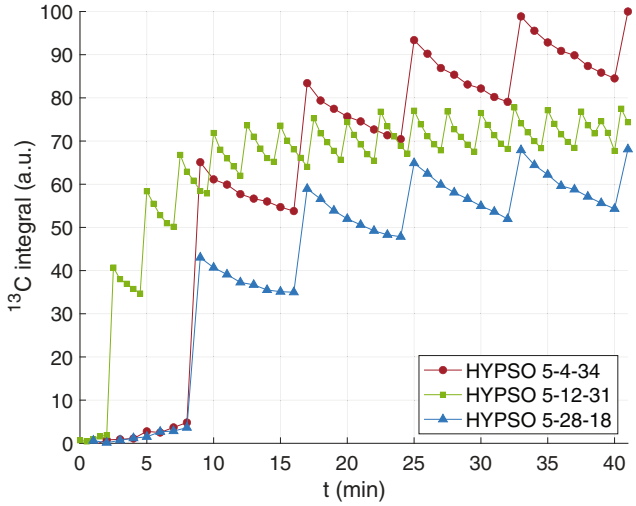


Figure 3. ^{13}C hyperpolarization (i.e., ^{13}C NMR signal intensity, normalized to the number of ^{13}C spins in each sample, presented on a relative scale) with multiple CP steps obtained with HYPSO 5-4 34 μmol g^{-1} (dark red circles, •), HYPSO 5-12 31 μmol g^{-1} (green squares, ■) and HYPSO 5-28 18 μmol g^{-1} (blue triangles, ▲). HYPSOs were impregnated with a 3 M sodium [1-^{13}C]-acetate in H_2O/D_2O (1/9 v/v) solution and polarized at 1.6 K and 7.05 T (^{1}H nuclear Larmor frequency 300.14 MHz, ^{13}C nuclear Larmor frequency 75.48 MHz) (see the Experimental Section). CP was applied at regular intervals (see the Experimental Section), and ^{13}C signals were acquired every 60 s (30 s in case of HYPSO 5-12 31 μmol g^{-1}) with a single transient (*rf*-pulse flip-angle = 4.4°).

porous volume unit) is almost the same and the pore size varies by a factor of 2.4, while the normalized ^{1}H signal integral drops only by a factor of 1.4. Interestingly, the build-up rate is seemingly independent of the size of the pores when the sample contains [1-^{13}C]-acetate, in contradiction with the previous results for HYPSO 5-12 and HYPSO 5-28 presented in Figure 1. This requires further investigation, but one possible explanation could be related to the presence of methyl groups, for example accelerating nuclear spin diffusion, or affecting electron spin relaxation and therefore DNP dynamics.

Finally, we show that such high and fast-building ^{1}H polarization can be readily transferred to the surrounding ^{13}C spins by CP methods following well-established protocols[3] currently used in our laboratory and to be soon incorporated in a commercial polarizer.

2.4. ^{13}C Polarization Transfer

Figure 3 shows ^{13}C polarization build-up curves with multiple CP steps (see Experimental Section). Note that the absolute solid-state ^{13}C polarization values are not presented, due to poor thermal equilibrium ^{13}C SNR, and will be confirmed in future studies. However, we observed from previous experiments in frozen DNP solutions that our protocol for multi-CP generally yields ^{13}C polarization reaching 70–90% of the available ^{1}H polarization.[16] Here the same or even higher efficiency can be anticipated since paramagnetic relaxation during CP is quasi-absent as radicals are located at the surface of the HYPSO matrix and therefore further away from the bulk nuclear spins within the pores.

Even though ^{13}C polarization was shown to decrease with the increased pore size (similar to ^{1}H polarization), we noticed an increase in the CP efficiency. Indeed, the CP relative efficiency factor (which does **not** represent the absolute CP efficiency, but allows the comparison of samples by comparing normalized ^{13}C integrals to normalized ^{1}H integrals (see Experimental Section for a detailed calculation procedure)) was found to be 0.88, 0.92, and 0.93 for HYPSOs 5-4-34, 5-12-31, and 5-28-18, respectively (see **Table 2** and Figure S1, Supporting Information for comparison of ^{1}H DNP and ^{13}C CP results and build-up curves). Since, in HYPSOs, the paramagnetic polarizing agents are located at the surface of the pores, an increase in the pore size leads to an increase in the average distance between the molecule and the radicals. Thus, the effects of paramagnetic interactions[27,28] (for example, shifts in nuclear spins' resonance frequencies) are reduced, preserving more ^{1}H spins available for $^1H \rightarrow {}^{13}C$ CP, and leading to an increase in overall CP efficiency.

For each HYPSO, the CP step delay was optimized to roughly correspond to the proton build-up time (8 min for HYPSOs 5-4-34 and 5-28-18, 2.5 min for HYPSO 5-12-31). Long proton build-up

Table 2. Comparison of ^{1}H DNP and ^{13}C CP results for three optimal HYPSOs: maximum ^{1}H NMR relative integral (normalized by the amount of ^{1}H nuclei), maximum ^{13}C NMR relative integral (normalized by the amount of ^{13}C nuclei), and CP relative efficiency factor (see the Experimental Section for the detailed calculation procedure). Note that presented values are comparable only within the same column.

Sample	^{1}H integral [a.u.]	^{13}C integral [a.u.]	CP relative efficiency factor [a.u.]
HYPSO 5-4-34	1	1	0.95
HYPSO 5-12-31	0.74	0.77	0.99
HYPSO 5-28-18	0.63	0.67	1

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times (see previous section) lead to longer CP inter-delays and therefore longer times to reach the ^{13}C polarization plateau. However, more than 80% of the final ^{13}C polarization can be achieved just in 17 min of multi-CP even for HYPSO 5-28-18.

3. Dissolution

Figure 4a shows a scheme of the experimental setup used for dissolution, filtration, transfer, and injection to a Bruker 600 MHz Ultrashield Plus spectrometer. The whole dissolution and transfer procedure was performed identically as for conventional frozen solutions, apart from a commercial Swagelock 2F series filter directly placed after the polarizer. This filter was chosen for the following reasons.

1) It features a 7 μm porosity of the sintered filtering element which leads to quantitative filtration of the HYPSO powder (particle sizes in the range from 25 to 40 μm were selected by sieving the synthesized solids), and the hyperpolarized solution purity was assessed with EPR spectroscopy (see the Experimental Section).

2) It does not lead to a measurable pressure drop and can therefore be placed inline without further modification of our setup or its dissolution and transfer parameters.

3) It does not lead to the creation of gas bubbles in the liquid plug during transfer, which enables the transfer and injection of a bubble-free liquid plug all the way to the NMR spectrometer, which is key for optimal performance of our rapid dissolution system.

4) It can be cleaned simply by flushing with gas (e.g., He or N_2), followed by 5 mL of water in the opposite direction to remove the residual particles from the filter.

The body of the filter and the sintered filtering element are made from nonmagnetic 316 stainless steel, which does not threaten the preservation of hyperpolarization. However, slightly magnetic 302 stainless steel is used for the spring which could screen the external magnetic field to some extent.[29] Traces of silicon-based lubricant initially used by the provider to fit in the filter may also be potentially present despite several washing cycles and might affect relaxation. To prevent possible zero-field crossings by the hyperpolarized solution, the filter was placed between an array of permanent magnets[30] providing a magnetic field of 5 mT.

Figure 4b shows the ^{13}C liquid state hyperpolarized signal decay of 1-^{13}C acetate after DNP using HYPSO 5-4-34, HYPSO 5-12-31, and HYPSO 5-28-18, after dissolution, filtration, and injection in a Bruker Ultrashield Plus 600 NMR spectrometer. The experiment was performed using 40 mg of each HYPSO impregnated with ≈30 μL of a 3 M [1-^{13}C]-acetate in H_2O/D_2O (1/9 v/v) solution and polarized with multi-CP at 1.6 K and 7.05 T (see the Experimental Section). After $P(^{13}C)$ reached a plateau, dissolution was performed into a Bruker Ultrashield Plus 600 spectrometer using a home-built system together with the inline commercial Swagelok filter (see the Experimental Section). ^{13}C signals were acquired every 5 s with single transients (*rf*-pulse flip-angle = 5°), Fourier transformed, and integrated. The decay was fitted with a monoexponential curve corrected for polarization depletion due to *rf*-pulses.[3] Obtained values of polarization and $T_1(^{13}C)$ relaxation times are presented in **Table 3**.

The fitted $T_1(^{13}C)$ were found to be 58.62 ± 0.05, 51.35 ± 0.05, and 58.4 ± 0.3 s for **HYPSO 5-4-34**, **HYPSO 5-12-31**, and **HYPSO 5-28-18**, respectively. As a comparison, the fitted $T_1(^{13}C) = 48.99 \pm 0.07$ s after dissolution of a standard 100 μL

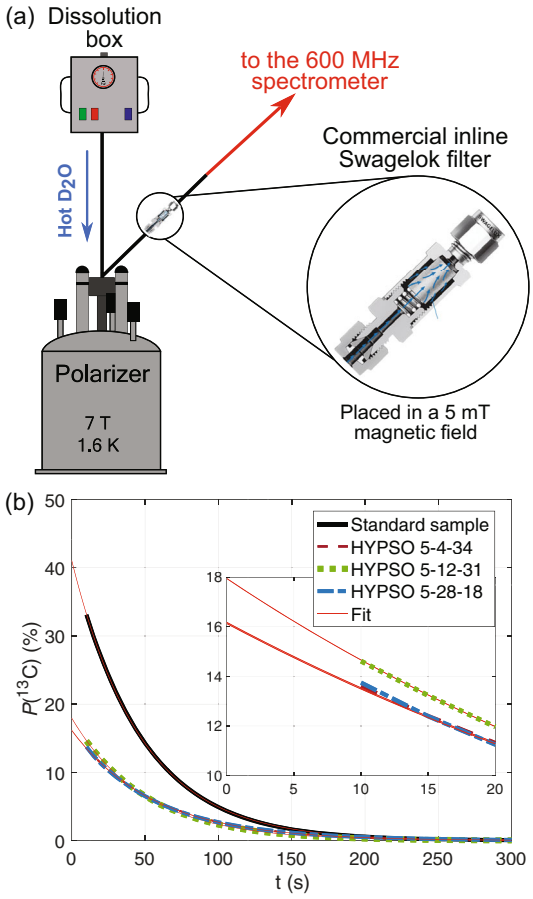


Figure 4. Filtered dissolution-DNP with HYPSO 5. After multi-CP, dissolution can take place "as usual" with a commercial filter that can be placed inline, thus leading to pure hyperpolarized solutions. a) Scheme of the setup used for dissolution, filtration, transfer, and injection based on a routine transfer setup with the sole addition of a commercial filter. b) ^{13}C liquid state hyperpolarized signal decay after DNP with HYPSO 5-4-34 (dark red dashed line), HYPSO 5-12-31 (green dotted line), and HYPSO 5-28-18 (blue dash-dotted line) dissolution, filtration, and injection in a Bruker Ultrashield Plus 600 spectrometer, along with exponential fit that takes into account polarization relaxation due to applied pulses (red line). The results for the standard formulation sample (after dissolution and injection) are also added for reference (black solid line). For the reference standard sample, 50 mM TEMPOL radicals were added to the 3 M [1-^{13}C]-acetate in H_2O/D_2O/glycerol-d_8 (1/3/6 v/v/v) solution. For HYPSOs, 40 mg of each HYPSO were impregnated with ≈30 μL of 3 M [1-^{13}C]-acetate in H_2O/D_2O (1/9 v/v) solution and polarized at 1.6 K and 7.05 T (see the Experimental Section). Several CP steps were applied (see the Experimental Section) until $P(^{13}C)$ reached a plateau, and dissolution was performed into a Bruker Ultrashield 600 Plus spectrometer. ^{13}C signals were acquired every 5 s with single transients (*rf*-pulse flip-angle = 5°). c) Comparison between hyperpolarized ^{13}C liquid state spectrum (red) of the first transient acquired 10 s after dissolution with 40 mg of HYPSO 5-12 31 μmol g^{-1} impregnated with 30.2 μL of a 3 M [1-^{13}C]-acetate solution, followed by filtration, transfer, and injection, and thermal equilibrium ^{13}C spectrum (blue), acquired after complete relaxation. Please note that the spectrum of hyperpolarized product was acquired with an *rf*-pulse flip-angle = 5°, while a 90° pulse was used for the thermal equilibrium.

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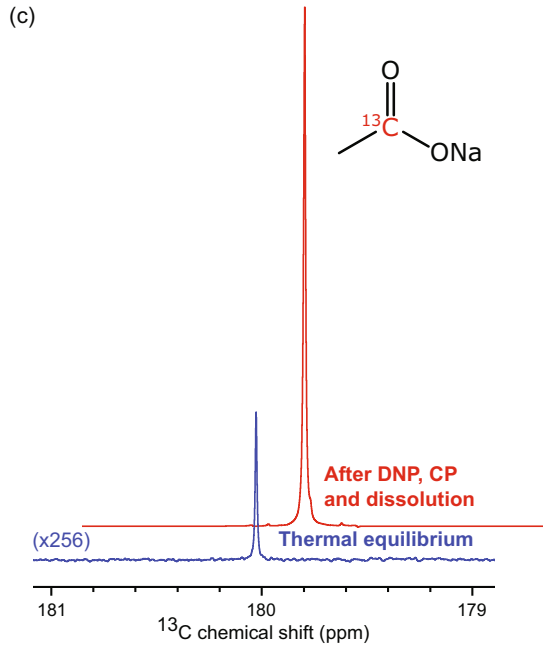


Figure 4. Continued.

Table 3. Comparison of the dDNP parameters for three optimal HYPSOs: maximum ^{13}C polarization in the liquid-state ($P_0(^{13}C)$), relative polarization loss during dissolution, filtration, and transfer (P loss, see Experimental Section for the detailed calculation procedure), and T_1 ^{13}C relaxation time.

Sample	$P_0(^{13}C)^{a)}$ [%]	P loss [a.u.]	$T_1(^{13}C)^{a)}$ [s]
Standard	41.05 ± 0.04	–	48.99 ± 0.07
HYPSO 5-4-34	16.018 ± 0.01	1	58.62 ± 0.05
HYPSO 5-12-31	17.95 ± 0.02	0.69	51.35 ± 0.05
HYPSO 5-28-18	16.13 ± 0.06	0.67	58.4 ± 0.3

[a)]Estimated from the exponential fit of the polarization decay curve. Please note that the presented errors correspond to the fitting parameters uncertainties.

3 M [1-^{13}C]-acetate in H_2O/D_2O/glycerol-d_8 (1/3/6 v/v/v) doped with 50 mM TEMPOL and dissolved with 7 mL of D_2O with the same parameters. While at a magnetic field of 14.1 T, there is no pronounced difference in the paramagnetic relaxation effect on T_1 between hyperpolarized samples originating from standard frozen solutions (and therefore containing mM concentrations of radicals) versus those originating from impregnated and filtered HYPSO (containing no measurable traces of radicals). However, we expect it to be more significant at lower fields, as shown by previous studies of radical scavenging,[31] and even more at ultralow fields[32,33] where paramagnetic relaxation is intensified. The EPR analysis of the HYPSO-hyperpolarized solutions did not reveal the presence of any paramagnetic species within a detection limit of 5 μM concentration (see Figure S2, Supporting Information, for comparison of EPR spectra of the filtered hyperpolarized solutions and TEMPOL solutions containing calibrated concentrations of nitroxide radicals). This represents an important step toward producing pure and directly injectable hyperpolarized solutions, eliminating the need for time-consuming purification. This is a decisive advantage when hyperpolarization vanishes with T_1 and is therefore hampered by prolonged purification methods.

Despite all the differences between the HYPSOs in the solid state, after dissolution and transfer, the ^{13}C polarization values in the liquid state were found to be roughly the same, although twice lower than the polarization achieved with the standard frozen sample formulation (see Table 3). Future improvements in the filter model or applied external magnetic field will likely increase polarization preservation.[29] Nevertheless, even with this rudimentary filtering setup, HYPSO 5 can provide very high signal enhancements without major modification of existing polarizers and could be used by the community as a standard polarizing agent. For example, Figure 4c shows the first hyperpolarized spectrum (same data as Figure 4b of [1-^{13}C]-acetate acquired 10 s after filtration, transfer, and injection (using HYPSO 5-12-31)). The calculated ^{13}C DNP enhancement was found to be ≈12,000-fold compared to thermal equilibrium at 14.1 T, with an outstanding signal-to-noise ratio of 82,000 measured with a 5° nutation angle pulse.

As expected, HYPSO 5-18-28, which displays the larger pores, demonstrated the lowest polarization loss during dissolution, filtration, and transfer (see Table 3). Compared to HYPSO with 4 nm pores at a radical concentration of 34 μmol cm^{-3}, HYPSO with 28 nm pores at 18 μmol cm^{-3} radical concentration increases polarization preservation during melting, filtration, and transfer, by a factor of 1.5. On the other hand, HYPSOs with larger pores and lower electron concentrations provide less final polarization in the liquid state.

Figure 5 shows the exchange rates (k) and intra/ex-pore T_1 rates (R) measured by T_2 REXSY on HYPSOs 5 with the smallest possible pore size (4-5 nm) and different radical concentrations (0, 14, and 72 μmol cm^{-3}). All the plotted maps and their associated methods section are presented in section 3 of Supporting Information. The combination of small pores together with high radical concentration inevitably leads to enhanced relaxation during dissolution step. However, even in the worst-case scenario, in the case of 1) smallest pores (4–5 nm), 2) highest radical concentration (72 instead of 34 μmol cm^{-3}), and 3) lower temperature

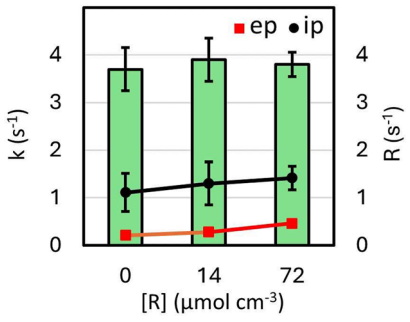


Figure 5. Exchange rates of molecules between the in-pore and ex-pore sites (k, green bars) due to molecular self-diffusion at room temperature are higher than the ex-pore (ep, red squares) and in-pore (ip, black circles) relaxation rates (R) for 3 M sodium [1-^{13}C]-acetate molecules in D_2O solution in HYPSO 5 solids (4–5 nm pore size) with different radical concentrations. The rates were determined by T_2-REXSY method as described in Supporting Information.

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(300 K instead of typ. 300–350 K during the dissolution process), the exchange of [1-^{13}C]-acetate between in-pores and ex-pores space occurs faster than nuclear spin-lattice relaxation. This makes it possible for the hyperpolarized solution to escape the pores upon dissolution at sufficiently high speed, therefore preserving at least part of its hyperpolarization.

4. Conclusion

We have demonstrated here that HYPSO 5 matrices can be synthesized with pore diameters of 12 and 28 nm, two to five times larger than those developed in previous studies.[9–12] We showed that these HYPSO 5 matrices can have their polarizing agent concentration tuned to an optimum for dDNP conditions. Additionally, we found that HYPSO 5 matrices with larger pores can be used for efficient DNP under dDNP conditions, while significantly reducing the concentration of paramagnetic polarizing agents and preserving high DNP performance. Finally, we showed that HYPSO 5 matrices can yield pure (i.e., not contaminated by radicals or glassing agent) hyperpolarized solutions without major modification of the dDNP setup and parameters. Indeed, DNP is performed with the exact same methodology as for standard frozen DNP solutions, and dissolution can be performed as usual without any modification but with the sole addition of a commercial in-line filter, and without noticeable effects on the fluid transport (no delay, pressure drop, or gas bubble creation). We are confident that with further optimizations, HYPSO 5 may compete with standard sample formulations and will be superior as they can provide pure and ready-to-inject hyperpolarized solutions for in vivo applications. In conclusion, HYPSO 5 could be readily used in any commercial dDNP polarizer and therefore has the potential to become a versatile polarization agent adopted by the dDNP community.

5. Experimental Section

Synthesis, HYPSO Characterization with EPR and N_2Adsorption, Particle Size Selection, Solution Purity by EPR

Cf Supporting Information.

DNP Sample Preparation

The samples used for ^{1}H DNP experiments (without CP and dissolution) were HYPSO 5 with two different pore sizes, 12 and 28 nm, and different concentrations of radicals (20, 31, 51, 67 and 9, 13, 18, 27 μmol cm^{-3}, respectively). Before ^{1}H DNP experiments, these samples were impregnated with a mixture of H_2O/D_2O (1/9 v/v). The volume of liquid for impregnation was calculated as $V_{liq} = 0.9 * V_p * m$, where V_p is the porous volume of HYPSO in cm^3 g^{-1} (as obtained from N_2 adsorption at $P/P_0 = 0.99$, see Table 1), and m is sample mass in g. This volume of solution corresponds to c.a. 90% of the HYPSO 5 total pore volume.

40 mg of each HYPSO 5 sample (with the following exceptions: 32 mg of HYPSO 5-12-20, 24.5 mg of HYPSO 5-12-51, 34 mg of HYPSO 5-28-27) were weighed in a glass cup and an appropriate volume of solution was added onto the powder by micropipette. With a glass rod, the sample was gently mixed for ≈1 min until a dry-looking homogeneous powder was obtained. The impregnated material was further introduced into the polarizer's home-built sample cup (made from fluorinated Kel-F polymer, and therefore with limited ^{1}H background signal from the cup itself) and directly inserted and frozen in the helium bath of the polarizer, precooled at 3.8 K, within less than 30 s.

For DNP experiments with polarization transfer to ^{13}C and subsequent dissolution, 3 samples were used: HYPSO 5-12-31, HYPSO 5-28-18, and HYPSO 5-4-34 (the latest is HYPSO 5 with smaller pore size for comparison, characterized in ref. [12]). 40 mg of each sample were impregnated with a 3 M solution of sodium [1-^{13}C]-acetate in H_2O/D_2O (1/9 v/v) mixture. To prepare the solution, 125.5 mg of sodium [1-^{13}C]-acetate purchased from Eurisotop (product number CLM-156-5) was simply diluted in 0.50 mL of H_2O/D_2O (1/9 v/v) mixture. The following preparation procedure was the same as above, albeit with a PEEK sample cup which made polarization quantification impossible due to its high ^{1}H content. Indeed, PEEK was used simply because it can withstand high pressures during dissolution while Kel-F can sometimes break.

In addition, a standard reference sample of 3 M sodium [1-^{13}C]-acetate in H_2O/D_2O/glycerol-d_8 (1/3/6 v/v/v), with 50 mM TEMPOL, was made following the standard procedure previously described.[3]

DNP Experiments

DNP experiments were carried out in a Bruker Biospin dDNP prototype polarizer operating at a field of 7.05 T (corresponding to a ^{1}H Larmor frequency of 300.14 MHz and ^{13}C nuclear Larmor frequency of 75.48 MHz) and temperatures down to 1.2 K. For DNP experiments with three best HYPSOs, a replacement pump with reduced pumping power was used which led to a base temperature of 1.6 instead of 1.2 K. A microwave source (Virginia Diodes Inc., Charlottesville, VA) operating in the range 196.8–198.8 GHz was used with an output power of ≈100 mW (at the output of the amplifier module), leading to 30 mW (measured by heating) at the sample. DNP was performed by irradiating with frequency-modulated microwaves[13] at a frequency centered around $f_{\mu w} = 197.660$ GHz using triangular frequency modulation of width $\Delta f_{\mu w} = 128$ MHz and a modulation rate $f_{mod} = 500$ Hz. For three best HYPSOs, ^{1}H NMR signals were measured with a 3D printed *rf* coil described elsewhere,[34] using an Avance II Bruker spectrometer with 64 transients *rf*-pulses with nutation angles of 0.1° applied every 5 s.

Cross Polarization (CP)

CP was used to transfer hyperpolarization from ^{1}H to ^{13}C nuclei at low temperature in the polarizer using the previously published protocol[3] involving microwave gating. In this particular case, the only parameter changed was the timing between CP pulses: for HYPSO 5-12-31 9 ms CP contacts were applied at 2.5 min intervals to allow the protons to significantly repolarize between contacts. For HYPSO 5-28-18, 8 ms CP contacts were applied every 8 min, and for HYPSO 5-4-34, 9 ms every 8 min.

Dissolution, Filtration, Transfer, and Injection

Dissolution was performed with 7 mL of D_2O preheated to about 170 °C at a pressure of 10 bars and pushed with 6 bars helium gas through a 1.5 mm i.d. PTFE capillary and a commercial Swagelok SS-2F-7 filter toward a home-built liquid driven transfer and injection system described elsewhere.[3,35] A hyperpolarized solution plug of 3 mL was selected in the transfer system and pushed over 4.8 m in less than 0.5 s toward a home-built injector placed in the Bruker Ultrashield Plus 600 NMR spectrometer equipped with a 5 mm

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CPQCI H/C/N cryoprobe. During the transfer, the solution was systematically kept at a minimal field of 5 mT by using a solenoid coil (0.5 mm diameter copper wire with 0.5 mm steps) wrapped around the 3.2 mm o.d. capillary, and driven with a current of 2.5 A, thus creating a field >5 mT. Two additional magnetic plates (see Figure S9, Supporting Information of the following article[30] for further details) were installed yielding a field >5 mT at the position of the transfer system valve and filter to counteract any field shielding effect of these metallic parts. The ^{13}C NMR signal acquisition started after 10 s setting time following injection and was repeated every 5 s with a single transient (*rf*-pulse flip-angle = 5°).

NMR/DNP Data Analysis

^{1}H build-up spectra were processed with a MATLAB (version R2023_a) script, written by Q. Stern (see "Data Availability" section). For experiments with best HYPSOs, ^{1}H and ^{13}C spectra were processed and integrated with TopSpin (version 4.3.0). Polarization values were calculated according to a procedure detailed elsewhere,[3] including corrections for *rf-pulse-induced* polarization depletion.

CP relative efficiency factor was calculated as follows. First, for each HYPSO, CP efficiency factor (CP_{eff}) was determined

$$CP_{\text{eff}} = \frac{\max(I^{\text{abs}}_{^{13}\text{C solid}}/n_{^{13}\text{C solid}})}{\max(I^{\text{abs}}_{^{1}\text{H solid}}/n_{^{1}\text{H solid}})} \quad (1)$$

where $I^{\text{abs}}_{^{13}\text{C solid}}$ ($I^{\text{abs}}_{^{1}\text{H solid}}$) are the absolute integrals of the ^{13}C (^{1}H) NMR signals in solid-state during CP (build-up), and $n_{^{13}\text{C solid}}$ ($n_{^{1}\text{H solid}}$) is the number of ^{13}C (^{1}H) nuclei in the sample. The CP efficiency factors for each HYPSO were then normalized to the highest CP efficiency factor.

The polarization loss during dissolution, filtration, and transfer (P_{loss}) was estimated as

$$P_{\text{loss}} = \frac{\max(I^{\text{abs}}_{^{13}\text{C solid}}/n_{^{13}\text{C solid}})}{\max(I^{\text{abs}}_{^{13}\text{C liquid DNP}}/I^{\text{abs}}_{^{13}\text{C liquid TE}})} \quad (2)$$

where $I^{\text{abs}}_{^{13}\text{C solid}}$ are the absolute integrals of the ^{13}C NMR signal in solid-state during CP, $n_{^{13}\text{C solid}}$ is the number of ^{13}C nuclei in the sample in the solid-state, $I^{\text{abs}}_{^{13}\text{C liquid DNP}}$ ($I^{\text{abs}}_{^{13}\text{C liquid TE}}$) are the absolute integrals of the ^{13}C NMR signal in liquid phase after dissolution (after complete relaxation to the thermal equilibrium). Note that $I^{\text{abs}}_{^{13}\text{C liquid DNP}}$ values were obtained from the fit of an experimentally obtained decay curve, to account for polarization losses due to first-scan delay and *rf*-pulses. The found P_{loss} values for each HYPSO were then normalized to the highest P_{loss}, yielding the relative polarization loss.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are openly available in [Zenodo] at [https://doi.org/10.5281/zenodo.13898049], reference number [13898049].

Keywords: dissolution dynamic nuclear polarization · hybrid polarizing solids · polarizing agents · silica · TEMPOL

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